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Discrepant Cretaceous paleomagnetic poles between Eastern China and Indochina: a consequence of the extrusion of Indochina

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Abstract

A paleomagnetic study of Cretaceous and Paleocene red-beds has been carried out in the Yunlong area (Yunnan province of China) of the Northern Indochina block. A high-temperature component with dual polarities was isolated from both Lower Cretaceous and Paleocene rocks, and of solely polarity for upper Early Cretaceous and lower Late Cretaceous rocks. Thirteen magnetic polarity zones are found along the Pijiang section. The primary nature of Cretaceous magnetic remanence is ascertained by positive and reversal fold tests. These results, combined with previous Cretaceous results from the Yunlong and Yongping areas of the Lanping basin, indicate that these areas behaved as a relatively rigid block, and rotated 37.2 ± 17° clockwise relative to Eurasia since the Paleocene. Comparing available Early Cretaceous paleomagnetic results obtained both from Indochina and Eastern China, a statistically significant paleolatitude difference of 9.6 ± 5.7° is inferred. This difference is consistent with an estimated 1000 ± 400 km extrusion of Indochina relative to South China, through a counter-clockwise rotation of 14° of Indochina around a finite rotation pole inferred from the stage poles of rotation derived from the magnetic isochrons in the South China Sea. The reconstruction is fully consistent between, the Cretaceous pole obtained from the Khorat basin of the stable core of the Indochina block and the coeval pole of Eastern China. © 2001 Elsevier Science B.V. All rights reserved.

Keywords: paleomagnetism; Cretaceous; Eastern China; Indochina; extrusion; India/Eurasia

1. Introduction

Continental deformation resulting from the collision and continued convergence between India and Eurasia is a matter of debate with crustal shortening and thickening and minor strike-slip faulting (England and Houseman, 1986; Dewey et al., 1989) on the one hand and lateral escape of blocks along major strike-slip zones (Tapponnier et al., 1982, 1990) on the other hand. Important aspects to test these models include determining the timing and nature of motion along the large-scale strike-slip faults that cut eastern Asia into blocks and the identifying rotations of these blocks. A series of precise dates including U–Pb monazite and $^{40}Ar/^{39}Ar$ K-feldspar ages on the Red River shear zone (Schärer et al., 1990; Harrison et al., 1996; Wang et al., 1998; Jolivet et al., 1999) reveals that ductile shear lasted between 35–17 Ma, which is coeval with the opening of the South China Sea (Briais et al., 1993).

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However, an intimate causal link between that zone and the opening of South China Sea still remains to be elucidated (Leloup et al., 1995).

Paleomagnetic studies in Indochina also provide constraints on the southeastward extrusion and clockwise rotation of the block (Yang and Besse, 1993; Yang et al., 1995). However, paleomagnetic results from northern Indochina imply complex deformation, e.g. large local rotations in this region, and sometimes, a minor southward displacement of the block (e.g. Huang and Opdyke, 1993).

In this paper, we present the Cretaceous and Paleocene paleomagnetic results from Yunlong area of west Yunnan province, in the northern part of Indochina, and then discuss their tectonic implications for continental extrusion along the Red River fault zone.

2. Paleomagnetic sampling and measurements

We sampled Cretaceous red-beds for a magnetostratigraphic study along the Pijiang river of Yunlong county (Fig. 1), in the northern Indochina block. The Cretaceous is divided into three formations, the Jingxing formation, Nanxin formation and Hutoushi formation. The Jingxing formation mainly consists of gray to green feldspathic sandstones in the lower part and siltstone and mudstone in the upper part. A rich Lamellibranchiata assemblage indicative of Early Cretaceous age was reported from the Jingxing formation (YNBGMR, 1996). The overlying Nanxin formation consists of red sandstone and siltstone and contains a rich Ostracoda and Lamellibranchiata fauna. The assigned age is late Early Cretaceous. The Hutoushi formation conformably overlies the Nanxin formation, which is composed of gray sandstones interbedded with mudstones. The assigned age is late Early Cretaceous or Late Cretaceous on the basis of regional stratigraphic correlation. It disconformably underlies the Paleocene Yunlong formation. The Yunlong formation is composed of red mudstone and siltstone, interbedded with mud-conglomerate, gray mudstone and marly limestone, and yields abundant Ostracoda and Estheria (YNBGMR, 1996). Twenty-seven samples from the Lower Cretaceous Jingxing formation (2 sites), 62 samples from the upper Lower Cretaceous Nanxin formation (7 sites), 16 samples from the lower Upper Cretaceous Hutoushi formation (2 sites) and 14 samples from the Paleocene Yunlong formation (2 sites) were collected using a portable drill. The combined sampling sites spanned up to 100 m of stratigraphic thickness and contained somewhat different bedding attitudes. The samples were oriented with a compass.

Remanent magnetization was measured using a 2G magnetometer, while stepwise thermal demagnetization was performed with Schpponstedt equipment. Results were analyzed using principal component analysis (Kirschvink, 1980) and Fisher statistics Fisher (1953).

3. Paleomagnetic results

Three kinds of demagnetization behavior were displayed, as single, two and three magnetic components. After removing a lower temperature component (<300°C), a higher temperature component is isolated between 570 and 690°C and decays to the origin (Fig. 2). A middle temperature component was separated between 300 and 500°C for several samples (Fig. 2a, c and e). The in situ mean direction of the lower temperature component, $D = 6.9°$, $I = 44.4°$ and $A_{95} = 10°$ ($n = 66$), is close to the present field (local dipole field direction $I = 44°$). A negative fold test at 95% confidence (McFadden, 1990) indicates a recent viscous remanent origin for this component. Only higher temperature components are defined as characteristic remanent magnetization (ChRM) for the samples. The maximum laboratory unblocking temperature of about 690°C suggests that hematite carries the remanence, which is consistent with the results of acquisition of isothermal remanent magnetization (IRM) and thermomagnetic experiments (Sato et al., 1999).

Both normal and reversed polarity ChRM are observed, with a northeast declination (downward inclination) and a nearly antipodal southwest direction (upward) after tilt correction from both the Early Cretaceous Jingxing and Paleocene Yunlong formations (Fig. 2, Table 1). The ChRM directions from both formations pass the reversal test (McFadden and McElhinny, 1990). Eight and five polarity zones are present in the Lower Cretaceous Jingxing and Paleocene Yunlong formations, respectively. Exclusively normal polarity is present within the late Early Cretaceous Nanxin and early Early Cretaceous Hutoushi formations (118–85 Ma, Harland et al.,

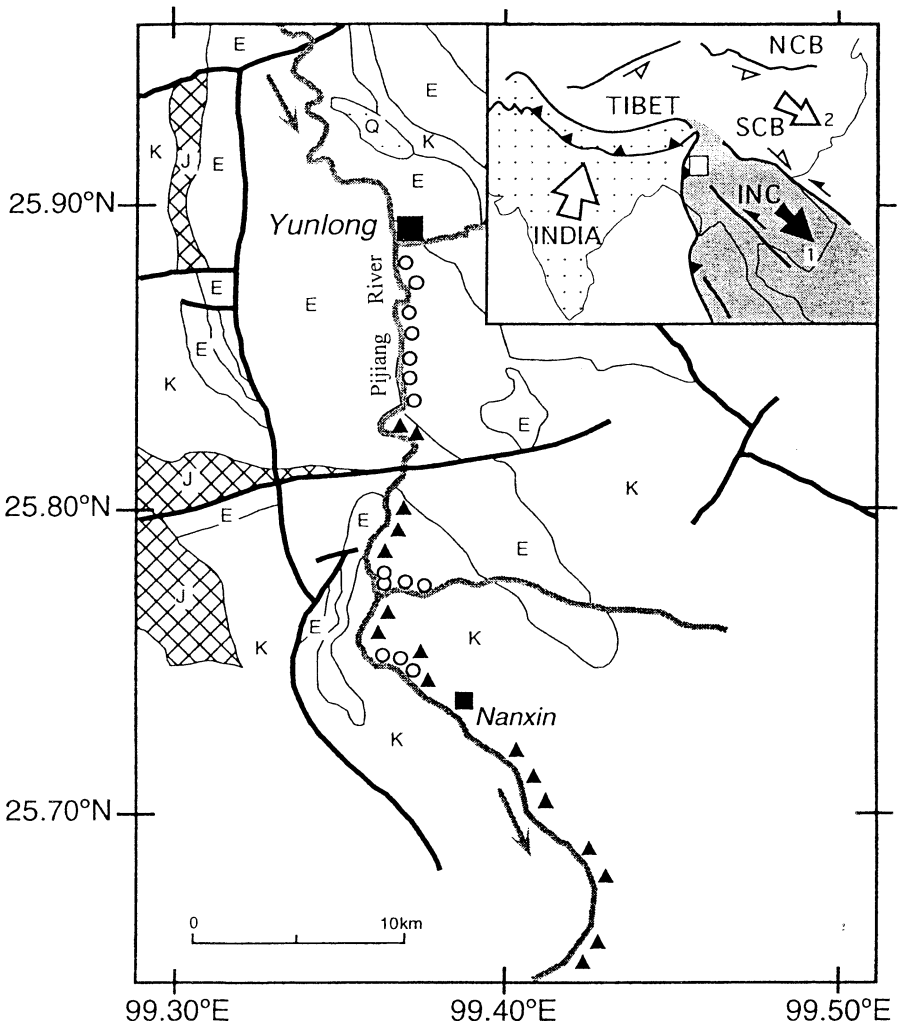


Fig. 1. Simplified geologic map of Yunlong area, western Yunnan (Indochina block), with sampling sites indicated as black triangles (open circles are from Sato et al., 1999). Inset map shows simplified large-scale Cenozoic extrusion in eastern Asia. Square shows the sampling region. Arrows correspond to the sense of strike-slip motion on major faults (solid for period before 20 My, open for age between 20–0 My) (Tapponnier et al., 1982).

1989), which is consistent with the Cretaceous Long Normal Superchron (CLNS) in the middle part of Cretaceous time. A positive fold test is achieved at the 95% confidence level ($f_{\text{in situ}} = 6.1$, $f_{\text{tilt corrected}} = 0.11 < f_{\text{critics}} = 3.5$, $N = 9$ sites, McFadden, 1990) for the Nanxin and Hutoushi formations. The folding event probably occurred in the Late Eocene (e.g. Leloup et al., 1995, as indicated by the disconformity between the mid-Upper Eocene continental molasse and Lower Eocene redbeds in the Jianchuan and Lanping basins (YNBGMR, 1996).

Upper Lower Cretaceous paleomagnetic results from the Nanxin formation around Yunlong have been reported earlier by Sato et al., (1999). Our sampling sites are distributed in between sites of Sato et al., (1999) (Fig. 1), that complemented the results for the Nanxin formation. New results from the Paleocene Yunlong, the early Late Cretaceous Hutoushi and Early Cretaceous Jingxing formations are added in this study. The mean direction of the Nanxin and Hutoushi formations from this study is statistically identical with 20 site-mean direction obtained by Sato et al.,

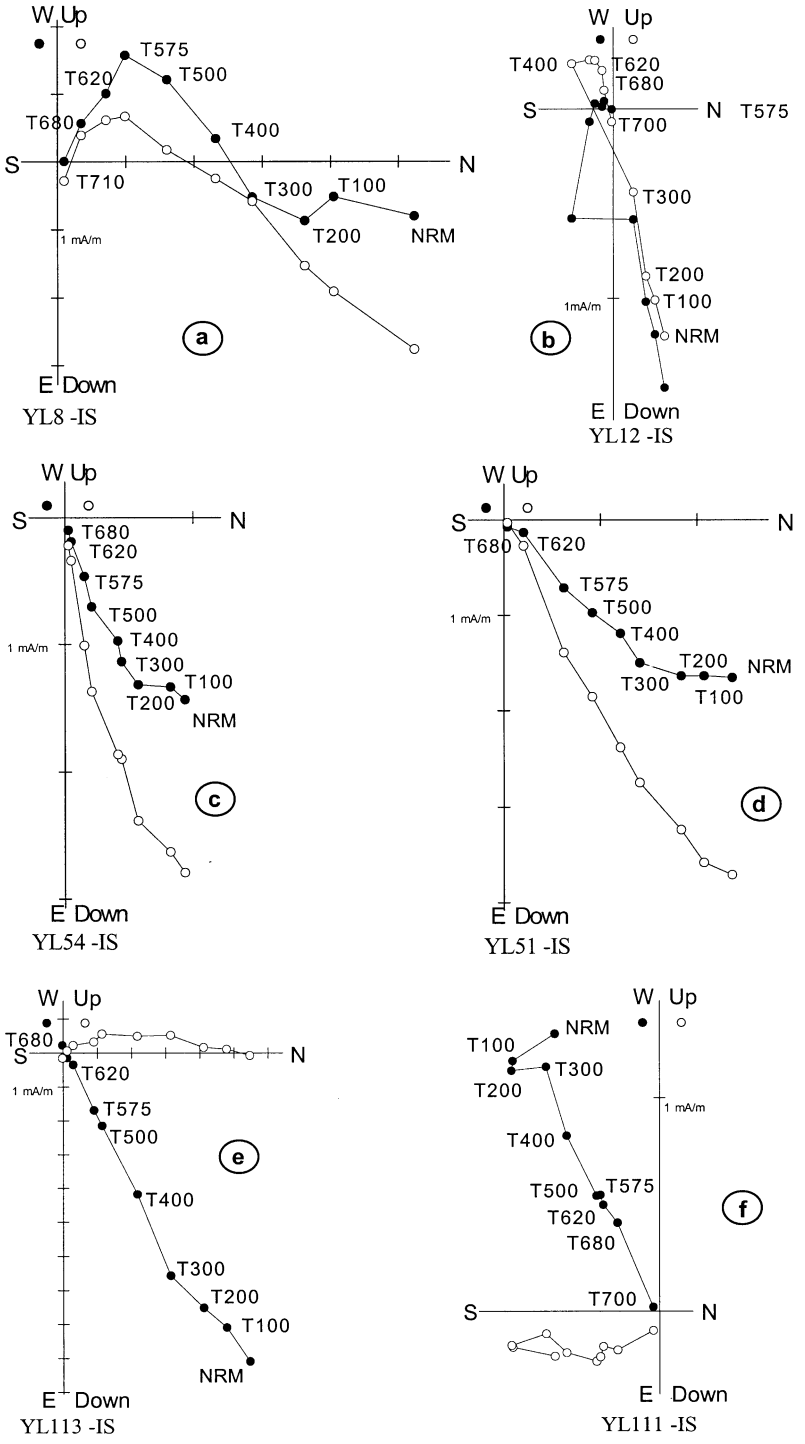


Fig. 2. Orthogonal vector projection of thermal demagnetization data of Early Cretaceous (a,b), late Early Cretaceous (c,d) and Paleocene (e,f) samples from Yunlong area. Solid (open) symbols refer to the projection on the horizontal (vertical) plane in geographic coordinates.

Table 1
Formation mean of paleomagnetic results from Yunlong area, Yunnan province (abbreviations: strike/dip = average strike azimuth and dip of each site; D/I ($k/\alpha 95$) = declination/inclination (precision parameter/half-angle of cone of 95% confidence about mean direction); subscripts 'g' and 's' represent geographic and stratigraphic coordinates, respectively; n/N: number of samples used/number of samples measured)

Site	Strike/dip	n/N	D_g	I_g	D_s	I_s	k_g/k_s	$\alpha_{95g}/\alpha_{95s}$
Jinxing formation (Lower Cretaceous)[a]								
y 1–14	273/39	12/14	106.6	39	74.5	37.1	4.9/7.0	21.8/17.7
y 15–27	306/44	11/13	124.8	78.2	52.5	44.7	2.7/10.2	35.1/15.0
Normal		17	96.1	57.5	56.99	38.9	3.3/11.3	23.3/11.1
Reversed		6	317.4	− 69.9	251.1	− 47.5	2.8/3.4	49.9/42.4
Mean		23	103.2	61.6	59.7	41	3.2/7.4	20.3/11.9
Nanxin and Hutoushi formations (Upper Lower Cretaceous)[b]								
y 28–36	167/49	8/9	55.3	6.4	41.7	50.7	46.0/59.4	8.3/7.2
y 37–45	133/40	8/9	36	15.4	31.7	54.5	8.7 /11.2	19.9/17.3
y 46–53	248/23	8/8	73.1	56.2	40.9	51.7	38.3/29.0	9.1/10.5
y 54–61	250/15	8/8	53.5	51.7	38.2	45.5	30.9/29.3	10.1/10.4
y 62–69	249/24	8/8	17	74.5	354.9	52.5	5.0/5.4	27.6/26.4
y 70–77	276/24	8/8	93.1	80.8	27.3	64.1	30.7/24.8	10.2/11.4
y 78–89	294/34	11/12	73	74.3	40.6	44.1	9.9/10.3	15.3/14.9
y 91–98	313/36	8/8	284.8	84.4	34.1	56.6	33.5/29.3	9.7/10.4
y 99–105	313/43	7/7	88.8	85.7	47	44.1	18.9/14.7	14.4/16.4
Mean		9	52.4	62.1	34	52.4	6.7/50.1	21.5/7.3
Yunlong formation (Paleocene)[c]								
y106-111	126/36	6/6	46.3	5.4	47.9	29.7	6.5/9.4	28.5/23.1.
y112-119	140/43	5/8	52.5	10.1	53	32.6	13.0/17.2	22.0/19.0
Normal		7	47	0.8	48.2	32.8	6.8/8.7	25.0/21.7
Reversed		4	232.8	18.3	233.3	28.2	30.9/41.8	16.8/14.4
Mean		11	49.2	7.6	50.2	31.1	9.1/12.9	16.0/13.2

[a] Reversal test (McFadden and McElhinny, 1990): angular distance 13.4, $\gamma = 29.0$; positive at 95% level.
[b] Fold test: $N = 9$; (1) McElhinny's method (1964), $k_s/k_g = 7.487 > F$ (14,16) = 3.37 at 99% level; (2) McFadden's method (1990), at 99% = 4.849, (in situ) = 6.078, (tilt corr.) = 0.11. Fold test: $N = 29$ ($N = 9$ from this study and $N = 20$ sites from Sato et al., (1999)); (1) McElhinny's method (1964), $k_s/k_g = 8.928 > F$ (54,56) = 1.875 at 99% level; (2) McFadden's method (1990), at 95% = 6.264, at 99% = 8.857 (in situ) = 22.25, (tilt corr.) = 3.939. Fold test is positive at 95 and 99% significant levels.
[c] Reversal test (McFadden and McElhinny, 1990): angular distance 6.3, $\gamma = 28.9$; positive at 95% level.

(1999). The fold test is positive at the 95% confidence level either using our Upper Cretaceous results, or combining these two results together (Fig. 3).

The pre-folding magnetization for the Nanxin and Hutoushi formations, and a positive reversal test for the directions from both the underlying Jingxing formation and overlying Yunlong formation, support that the characteristic directions from these formations are primary remanences.

4. Tectonic implications

Because of the limited number of samples, we emphasize that the results obtained from the Lower Cretaceous Jingxing and Paleocene Yunlong formations are preliminary. However, it is clear that Yunlong rotated $37.2 \pm 17°$ clockwise relative to Eurasia (Besse and Courtillot, 1991) since the Paleocene, calculated by the statistical method of Butler (1992). The Late Early Cretaceous paleopole calculated from 29 sites, including our data and that of Sato et al., (1999), is located at 56.7°N, 170.1°E with $A_{95} = 4.0°$. This pole (pole 2 in Fig. 4) is indistinguishable from a coeval paleopole obtained from the Yongping area (Funahara et al., 1993b) (pole 3 in Fig. 4), indicating that either the Yongping and Yunlong areas behaved as a rigid block relative to the Simao basin, or that these areas both rotated in the same sense, about $7 \pm 5.5°$ clockwise with respect to the Khorat basin within the Indochina block.

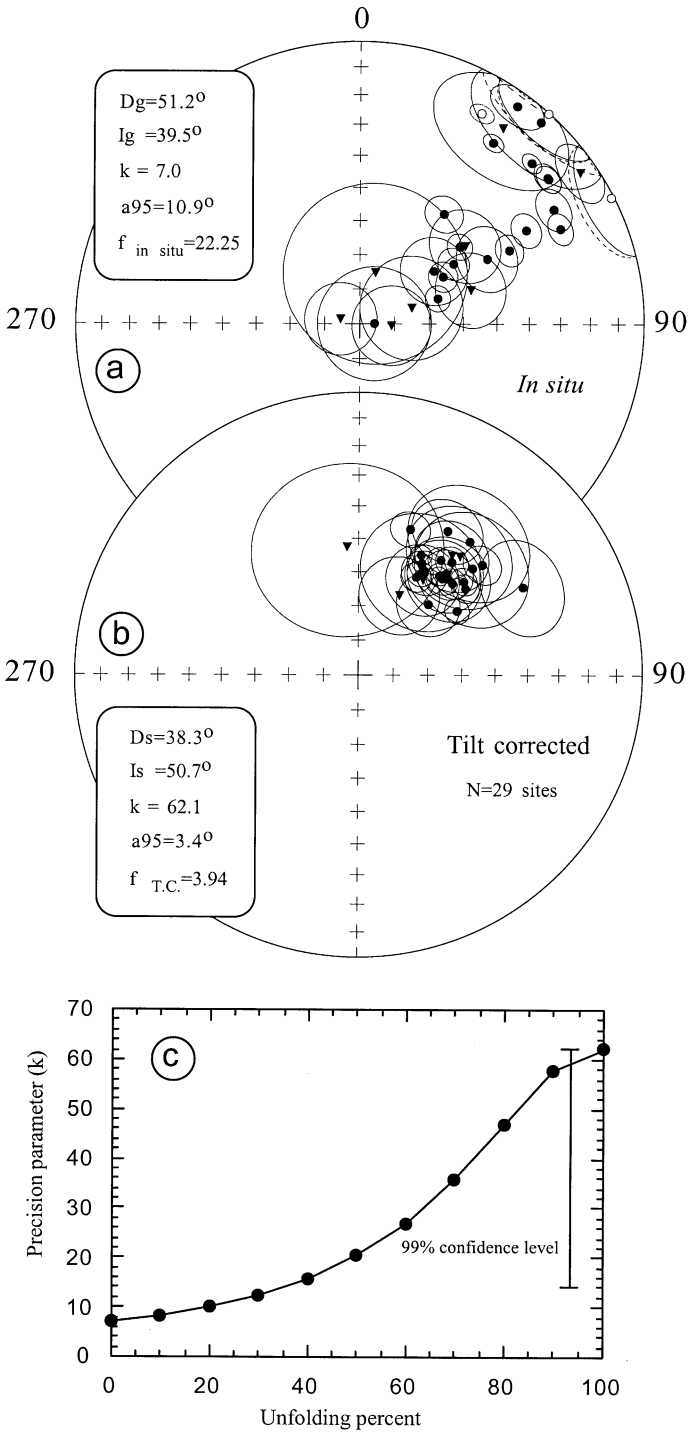


Fig. 3. Equal-area projection of the late Early Cretaceous/early Late Cretaceous site-mean ChRM directions of Yunlong area before (a) and after (b) tectonic corrections. Solid (open) symbols represent directions plotted onto the lower (upper) hemisphere. Triangle and circle indicate site-mean direction from this study and Sato et al. (1999), respectively. A positive fold test is achieved at the 95% confidence level ($f_{\text{critics}} = 6.264$ for $N = 29$ sites, McFadden, 1990). (c) Progressive unfolding of the site-mean direction shows a maximum concentration at 100% unfolding.

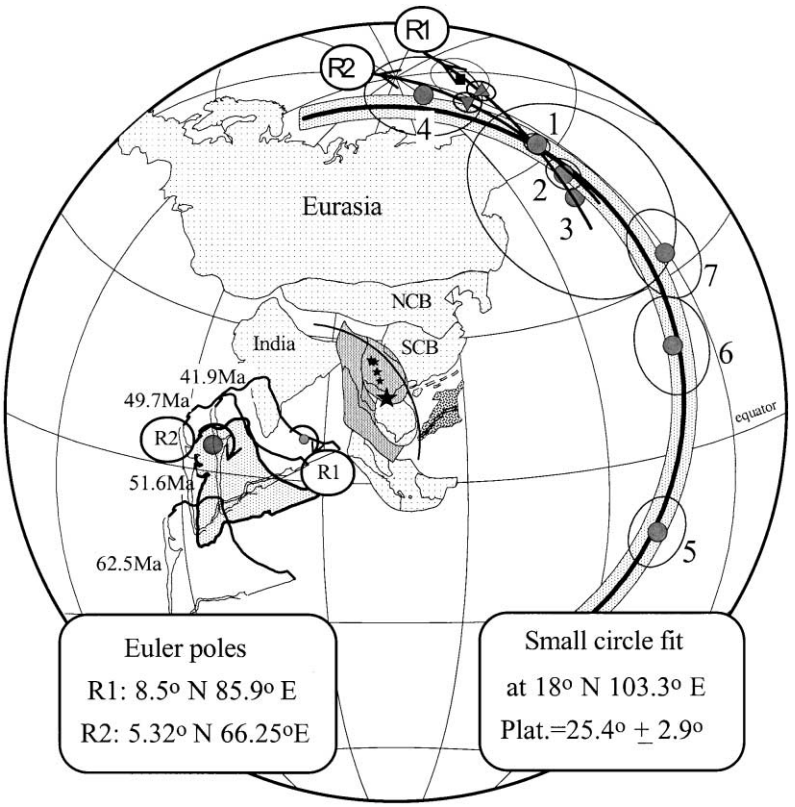


Fig. 4. Comparison of late Early Cretaceous–early Upper Cretaceous paleopoles between Eastern China (solid squares) and Indochina (solid circles). Note that the distribution of poles for Indochina is clearly along a small circle centered in the sampling areas (stars). Numbers next to the pole are the same as in Table 2. The fit of the small circle centered at reference point 18°N and 103.3°E (northern Khorat basin) gives a colatitude of 64.6 ± 2.9°. Indochina block is restored by rotating of 14° around the finite pole R2 at 5.32°N/66.25°E. Paleopole of Indochina is also rotated around R1 and R2, shown as upward and downward triangles, respectively. Paleopositions of India relative to present Eurasia during Early Tertiary time are also given (Patriat and Acharche, 1984) indicative of vast extruded space between India and Eurasia.

Table 2
Cretaceous paleomagnetic results from the Indochina block (Nr: numbers denoting poles in Fig. 4. Test: R = reversal test, F = fold test; Age: see text for explanation; *N*: number of sites)

Nr	Basin	Location (°N/°E)	Age	*N*	$D(°)/I(°)/\alpha_{95}$	λ(°N)/ϕ(°E)	A_{95} (dp/dm)	Test	References[a]
1	Khorat	16.5/103.0	K2-2	10	28.1/40.5/2.4	62.7/173.3	2.4		1
2	Lanping	25.8/99.4	K2-2	9	34.0/52.4/7.3	69.7/167.6	6.9/10	F	this study
			K2-2	20	40.2/49.9/3.9	54.6/171.3	4.4	F	2
		(Mean)	K2-2	29		56.7/170.1	4.0	F	this study
3	Lanping	25.5/99.5	K2-2	12	42.0/51.1/15.7	50.9/167.3	20.6	F	3
			(Bivariate Mean)		40.1/49.7/16.9/7.4	54.4/172.0			
4	Lanping	25.6/100.2	K2-2	9	6.9/47.7/8.6	83.6/152.7	10.0	R	4
5	Simao	23.5/100.7	K1-2	11	110.4/36.8/5.3	9.2/163.5	5.5	R.F	5
6	Simao	23.4/100.9	K2-2	8	79.4/43.3/9.1	18.9/170.0	8.9		4
7	Simao	21.6/101.4	K2-2	10	60.8/37.8/7.6	33.7/179.3	8.2		4

[a] References: 1: Yang and Besse, 1993; 2: Sato et al., 1999; 3:Funahara et al., 1993a,b; 4: Huang and Opdyke, 1993; 5. Chen et al., 1995

The available Indochina block Cretaceous poles (Table 2) are distributed along a small circle centered on their sampling areas (Fig. 4). The small circle that passed over the poles is fit at reference point 18°N and 103.3°E (northern Khorat basin) with a colatitude of 64.6 ± 2.9°. Relative rotations between these areas are significant, which could be the result of non-rigid behavior of the block in the northern part of Indochina. We note that, except for the Early to late Early Cretaceous paleopoles of Chen et al. (1995) and Funahara et al. (1993a), the rest of the Cretaceous data are dominated by normal polarity, which is consistent with magnetization acquired during the CLNS. We therefore prefer the interpretation that the paleolatitude of Indochina at the reference point was 25.4 ± 2.9° during the CLNS (118–84 Ma).

The Cretaceous is generally divided into Early and Late epochs, which spans about 80 Ma between 145 and 65 Ma (Harland et al., 1989). The Cretaceous magnetic time scale, however, shows three divisions, with the CLNS in the middle, and mixed polarities in the Early (between 145 and 118 Ma) and Late Cretaceous (84–65 Ma). Cretaceous strata from the North China Block (NCB) and South China Block (SCB) are generally dated by paleontology. Thus, if magnetic polarity can be well determined, then the strata could correspond to the time of these three magnetic epochs, early Early Cretaceous (referred to as K1-1 from 145 to 118 Ma), the CLNS (referred to as K2-2 between 118 and 84 Ma), or the late Late Cretaceous (referred to as K3-3 between 84 and 65 Ma). The average paleomagnetic poles of these three epochs are more suitable for the aim of paleogeographic reconstruction. Cretaceous paleopoles of the NCB and SCB were derived from compilations of Yang et al. (1995) (Table 3). Since then, several additional Cretaceous paleopoles have appeared. For instance, Gilder and Courtillot (1997) and Gilder et al. (1999) published a series of new Early Lower Cretaceous poles from Anhui and Shandong provinces of the NCB, which pass the fold and/or reversal tests. A Late Cretaceous pole to Early Tertiary was also obtained from Anhui province of the SCB (Gilder et al., 1999). Another coeval pole obtained from the Ordos basin and Alashan areas of the NCB features a positive reversal test (Wu et al., 1993). The K1-1 mean pole for the NCB is at 76.4°N and 209.5°E ($A_{95} = 4.3°$).

Significant amounts of Cretaceous poles come from the cratonic margin in the SCB, e.g. Sichuan, Yunnan, Guangxi, Guangdong, Hainan, Anhui and Fujian provinces (Enkin et al., 1991; Gilder et al., 1993; Otofuji et al., 1998; Gilder et al., 1999). Some poles (Enkin et al., 1991; Gilder et al., 1993; Funahara et al., 1993a; Li et al., 1995) have clearly suffered local rotations about vertical axes as suggested by the original authors. These poles are clearly dispersed along a small circle centered on the sampling areas (Fig. 5(b)). Instead of computing a mean pole using classical Fisherian statistics, a method which estimates the mean pole by a combination of pole directions and small circles was used (Besse and Courtillot, 1991). The poles, which are suspected to come from areas rotated about vertical axes, are thus displaced by small circles the distance of which to the sampling sites is the paleocolatitude. By iteration, we then find a pole position with a Fisherian average of the pole directions and the nearest points on the small circles closest to this pole position. The method, described by McFadden and McElhinny (1988) estimates the semicircle of confidence. Endpoints after iteration are shown in Fig. 5 as solid triangles. Because the offset of Triassic granitic plutons lined along the Xian-Shui-He fault was estimated as 100 km (Mattauer et al., 1992), paleomagnetic results from rocks west of the Xian-Shui-He and An-Ning-He fault zone were also taken into account, e.g. Tamai et al. (1996); Otofuji et al. (1998); Funahara et al. (1993a,b); Zhu et al. (1988); Huang and Opdyke (1992). Except for the result reported by Zhu et al., (1988), the other results all pass the fold test. The K1-1 mean pole for the SCB stands at 78.9°N and 214.9°E ($A_{95} = 8.1°$). The early Early Cretaceous mean pole of the NCB is indistinguishable from that of the SCB and Eurasia at 95% confidence (Fig. 5(c) and (d)), which supports the interpretation that the NCB and SCB were sutured together prior to the Middle Jurassic (Yang et al., 1992) or the Late Jurassic (Gilder and Courtillot, 1997). The K2-2 mean pole characterized by normal polarity, is insignificant from the Early Cretaceous pole, however, located farther east (Fig. 5(c)) and coincident with the 80–110 Ma poles of Eurasia (Besse and Courtillot, 1991). The Late Cretaceous pole is also consistent with the 60–80 Ma poles of Eurasia, although with larger uncertainty (Fig. 5(a)).

Table 3
Cretaceous poles from the North and South China Blocks

Block	Site location (N/E)		Pole position (N/E)		A_{95} (dp/dm)	References
Late Cretaceous						
NCB	40.1	112.9	79.6	170.1	5.8	Zheng et al. (1990)
SCB	22.2	114.2	78.2	171.9	10.6	Chan et al. (1991)
SCB	25	116.4	67.9	186.2	9.2	Hu et al. (1991)
SCB	26	117.2	66.9	221.4	5.4	Zhai et al. (1992)
SCB	29.1	104.6	71.8	227.3	7	Enkin et al. (1992)
SCB[a]	30	102.9	74.8	250.8	6	Enkin et al. (1991)
SCB	23	115	66	221.5	2.3/4.2	Hsu (1987)
SCB[a]	30.8	118.2	83.8	200.3	14.6	Gilder et al. (1999)
Mean K3 $N=8$			75.1	210.3	6.8	Fisher's statistics
			72.6	208.0	6.5	Small circles intersection
Cretaceous Long Normal (CLN)						
SCB[a]	26.6	102.4	78.9	186.6	5.5	Zhu et al. (1988)
SCB	32	119	76.3	172.6	10.3	Kent et al. (1986)
SCB[a]	26.5	102.4	81.9	220.9	7.1	Huang and Opdyke (1992)
SCB[a]	27.9	102.3	84.8	245.8	5.7	Tamai et al. (1996)
SCB[a]	25.9	101.7	64.6	199.6	3.3	Otofuji et al. (1998)
SCB[a]	19	109.3	83.2	143	9.8	Li et al. (1995)
SCB[a]	22.2	108.7	86.5	26.4	10	Gilder et al. (1993)
SCB[a]	23.1	113.3	56.2	211.5	3.9	Gilder et al. (1993)
SCB[a]	26	117.3	65.1	207.2	5	Gilder et al. (1993)
Mean CLN $N=9$			76.9	200.5	8.3	Fisher's statistics
			76.4	200.4	4.9	Small circles intersection
Early Cretaceous						
NCB	35	108	75.8	208.7	7.5	Ma et al. (1993)
Mongolia	45.4	107.6	72.5	205.3	7.6	Pruner (1992)
Mongolia	42	119.2	82.9	249.5	5.7	Zhao et al. (1990)
NCB	37.2	105	74.5	203.4	8.2	Wu et al. (1993)
NCB	31.6	116	74.5	201	4.7	Gilder and Courtillot (1997)
NCB[a]	36.9	120.7	81.5	218.9	4.8/6.8	Gilder et al. (1999)
SCB	29.7	120.3	77.1	227.6	5.5	Lin et al. (1985)
SCB[a]	30	102.9	74.5	229	3.2	Enkin et al. (1991)
SCB[a]	30	103	78.6	201	15.3	Zhuang et al. (1988)
SCB[a]	26.8	102.5	69	204	4.3	Huang and Opdyke (1992)
SCB[a]	25	101.5	49.2	178	11.4	Funahara et al. (1993a,b)
Mean						
NCB $N=6$			76.4	209.5	4.3	Small circles intersection
SCB $N=5$			78.9	214.9	8.1	Small circles intersection
NCB + SCB $N=11$			76.8	210.2	3.1	Small circle intersection
Eurasia 90 Ma			76.3	200.9	3.2	Besse and Courtillot(1991)
Eurasia 100 Ma			76.0	198.7	5.0	Besse and Courtillot(1991)

[a] Poles suspected suffering local rotation.

There is discordance between the K2-2 pole of Eurasia (or Eastern China) and the colatitude small circle band for Indochina (Fig. 5d). The latitudinal difference between Eurasia and Indochina is $9.6 \pm 5.7°$ (or $9.2 \pm 5.8°$ between Eastern China and Indochina) at the reference point of 18°N and 103.3°E, which gives a more precise estimation, comparing with that of Yang and Besse (1993) or Sato et al. (1999). This difference implies a significant southward motion of Indochina relative to Eastern China since the Late Cretaceous.

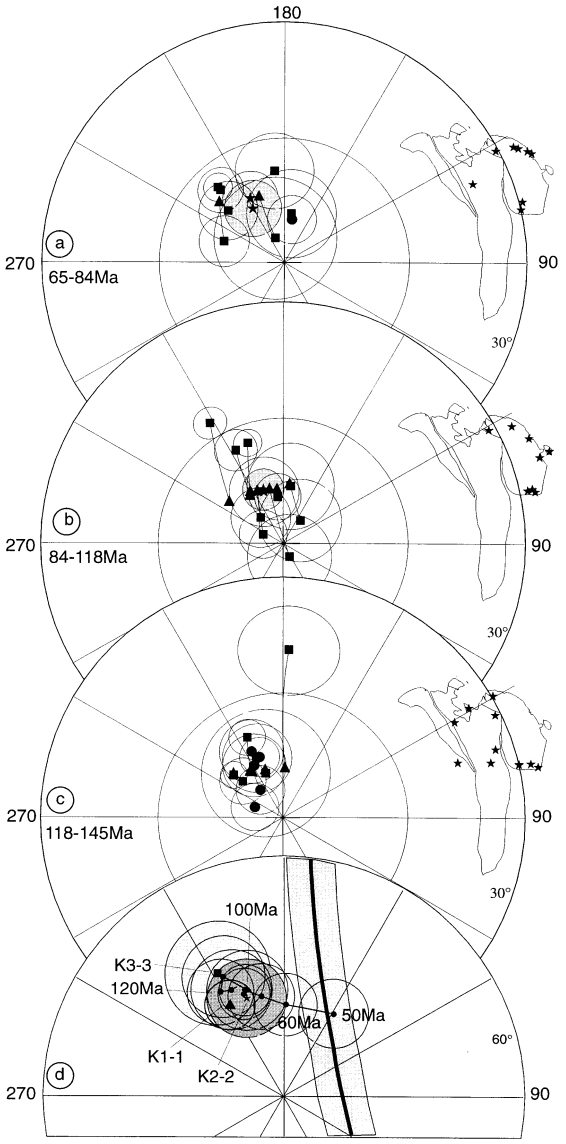


Fig. 5. Equal-area projections of (a) late Late Cretaceous (84–65 Ma), (b) late Early Cretaceous (118–85 Ma) and (c) early Early Cretaceous (145–118 Ma) paleopoles from the North and South China blocks with their 95% half-angle of confidence. Paleopoles suffered local rotations and after iteration are shown as circles (North China)/or squares (South China) and filled triangles, respectively. The mean pole position is indicated by a star with a 95% half-angle of confidence. (d) Comparison between the synthetic Eurasian paleopoles from 50 to 130 Ma (Besse and Courtillot, 1991) and mean K1-1, K2-2 and K3-3 poles of Eastern China, the small circle centered at Khorat basin (18°N and 103.3°E) is also shown.

Independently, using computed fits of magnetic isochrons (anomaly 5–11) from the South China Sea, Briais et al. (1993) inferred two poles of finite rotation of Indochina relative to South China. One pole, calculated from cumulative stage pole rotations derived from the magnetic isochrons (Table 3 in Briais et al., 1993), is located at 8.5°N and 85.9°E (referred to as R1), and another is an average of the stage poles, weighted by the duration of each stage, located at 5.32°N and 66.25°E (referred to as R2). Using these poles, the opening of the South China Sea results in about 12.3° of clockwise rotation of Indochina relative to South China.

For the reference point at 18°N and 103.3°E, the declination difference between the Khorat basin and Eastern China is 14.4 ± 5.7°, based on K2-2 poles, or 13.9 ± 5.8° between the Khorat basin and Eurasia. Both of them are consistent with the kinematic model of the opening of the South China Sea. We restore the Indochina block using the poles of finite rotation inferred by Briais et al. (1993) (1993), 14° around finite rotation poles R1 and R2, respectively. Using R1 results in a discordance between the CLNS pole obtained from the Khorat basin and the reference pole of Eurasia (with difference 6.2 ± 5.4°) or Eastern China (with difference 5.9 ± 5.5°). When the finite rotation pole, R2, is used for the same procedure, the rotated pole of the Khorat basin is identical to those coeval poles of Eastern China and Eurasia, being 2.6 ± 5.4° and 2.6 ± 5.5° respectively. Thus, the finite rotation pole R2 gives a best estimate for the paleomagnetic data.

The rotation (13.9 ± 5.8°) of Indochina relative to Eurasia inferred from paleomagnetic data is indistinguishable from that derived by Briais et al. (1993) from the fits of magnetic isochrons (An5–An11). By rotating 14° of a reference point (21.5°N/105°E) along the fault trace around the finite rotation pole R2, we estimate a minimum southeastward displacement of Indochina to be 1000 ± 400 km along the Red River Shear zone that initiated probably in the Paleocene as indicated by the 50–60 Ma ages found by Zhang (1995) in the North of the Red River fault zone. This offset along the Red River fault is larger (although compatible within the uncertainty) than that deduced by the kinematics of the South China Sea and geological markers proposed by Leloup et al. (1995). This is also consistent with dislocation of

Shiwandashan-Gan-Hang zone defined by the granites enriched in Nd and Sm in the SCB (Gilder et al., 1996). A larger amount of displacement of Indochina would require an earlier phase of extrusion (Yang et al., 1995; Zhang, 1995), taking into account the Paleogene crustal extension along the South China Sea margin prior to seafloor spreading, e.g. the Paleogene Bei-bu-wan, Qiong-dong-Nan and Zhujiang-kou basins, and the more than 100 km of right-lateral displacement on the Red River fault since the Pliocene (Allen et al., 1984; Yang and Besse, 1993). This displacement amount of Indochina is consistent with that deduced from paleolatitudinal differences ($9.7 \pm 4.3°$ or about 1500 ± 600 km displacement). This consistency, constrained by both paleomagnetic data and kinematics of seafloor spreading in the South China Sea, links the southeastward displacement of Indochina relative to South China to the opening of the South China Sea.

5. Conclusion

We added new paleomagnetic results from the Paleocene Yunlong, late Early Cretaceous Hutoushi and Lower Cretaceous Jingxing formations to those of Sato et al. (1999). The pre-folding magnetization obtained from 29 sites in the Nanxin and Hutoushi formations, and a positive reversal test for the directions from both the underlying Jingxing formation and overlying Yunlong formation, support that the characteristic directions from these formations are primary remanences. The paleolatitudinal difference of $9.6 \pm 5.7°$ between Indochina and Eurasia (or $9.2 \pm 5.8°$ between Indochina and Eastern China) is found. A clockwise rotation of $14.4 \pm 5.7°$ of Indochina relative to Eurasia (or $13.9 \pm 5.8°$ relative to Eastern China) is constrained by our paleomagnetic study and is in agreement with the estimates given by Tapponnier et al. (1982) and Briais et al. (1993). Sinistral motion on the Red River fault on the order of 1000 ± 400 km can be estimated by a rotation of Indochina around an Euler pole inferred from kinematic model in the South China sea, which gives a better estimate than the offset deduced from paleolatitudinal differences (1500 ± 600 km, Yang and Besse, 1993). This study strengthens the causal link between the extrusion of Indochina and the opening of the South China Sea (Tapponnier et al., 1982; Briais et al., 1993; Leloup et al., 1995.

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